

Behavioral Anomaly–Driven Risk Assessment for Access Control in Enterprise Systems

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Abstract—Insider threats are a major problem for enterprise security because harmful or unauthorized actions may be carried out by legitimate users, and their behavior can look similar to ordinary work activity. Traditional role-based access control (RBAC) decides access mainly from fixed roles and permissions, without accounting for changes in user behavior over time. As a result, it may not respond well when risk conditions shift during normal system use. This paper proposes a behavior-based risk assessment framework that combines unsupervised anomaly detection with access control decisions. The Study uses authentication and file activity logs from the CERT Insider Threat Dataset and grouped user actions into sessions based on time gaps between consecutive events. In this study, each session is described through behavioral features that reflect timing patterns, overall activity level, categories of file operations, use of removable media, and access outside normal working hours. An Isolation Forest is used to flag unusual sessions without labeled data, assign each session an anomaly score, and translates those scores into risk levels that are easier to interpret. To assess whether the identified risk matters in practice, a policy-based analysis is run against the organization’s predefined enterprise security rules. The experimental results show a clear association between high-risk sessions and security policy violations, whereas low-risk sessions show few or no violations. A comparative evaluation shows that static RBAC does not adequately limit risky actions, while the risk-aware approach identifies high-risk sessions and applies restrictions or monitoring only when needed. The results suggest that including behavioral risk in access control decisions can help reduce the risk of insider threats.

Keywords— Behavioral Analytics, Insider Threat Detection, Risk-Based Access Control, Isolation Forest, Unsupervised Learning, Enterprise Security.

I. INTRODUCTION

Insider threats continue to be a difficult security issue in modern enterprise settings [1]. In contrast to external attackers, insiders act using valid credentials and permitted access, so harmful or unauthorized actions can appear similar to normal day-to-day activity. These threats can come from deliberate attacks, careless behavior, or stolen accounts, which makes them hard to identify with traditional perimeter-based security controls.

As organizations rely more on digital systems to store and manage sensitive data and resources, it becomes necessary to detect and respond to insider behavior that may pose security and operational risks. Traditional access control systems,

especially role-based access control (RBAC), assign permissions mainly by fixed role definitions and static policy rules. RBAC can make it easier to manage authorization rules, but it does not capture what a user does after access has been granted. As a result, users who show abnormal or risky behavior may keep the same access rights as long as their assigned role does not change.

This limitation makes RBAC less suitable in dynamic settings, where security risks can change quickly in response to user behavior patterns rather than staying aligned with predefined roles. To reduce these limitations, recent studies have examined user and entity behavior analytics (UEBA) methods that review activity logs to detect unusual behavior [2]. Many methods aim to spot unusual patterns in login activity or network traffic, and they often depend on supervised learning models that need labeled examples of attacks. Reliable ground-truth labels for insider threats are difficult to obtain in practice because real incidents are rare, depend heavily on context, and are hard to label consistently.

This has led researchers to use unsupervised learning methods that learn patterns of normal behavior and then flag departures from those patterns without relying on labeled data [3]. This study presents a behavior-based risk assessment framework that links unsupervised anomaly detection with access control decisions. Instead of treating events as separate items, user activity logs are grouped into sessions that reflect continuous periods of interaction. Each session is described with behavioral features that capture timing patterns, overall activity level, types of file operations, use of removable media, and access during non-working hours. We then apply an Isolation Forest model to flag sessions as anomalous when they depart from typical behavioral patterns.

A central contribution of this study is to link anomaly detection results to concrete security decisions in real settings. Rather than treating anomaly scores as raw values, the observed behavioral risk is translated into clear risk levels and then checked against predefined enterprise security policies. This policy-based validation provides empirical evidence that sessions labeled as high risk align closely with behaviors commonly viewed as unsafe or unauthorized in enterprise settings, whereas low-risk sessions show little to no policy noncompliance.

The proposed framework is then assessed in terms of access control by contrasting static RBAC decisions with risk-aware decision making. The results show that static RBAC does not adequately limit risky behavior, while adding behavioral risk allows the system to restrict or monitor only those sessions that present a higher risk. This study uses

session-level behavior, unsupervised learning, and policy-based validation to support a practical, defensible method for strengthening access control systems against insider threats.

II. LITERATURE SURVEY

Insider threat detection has become a more common focus in recent years because perimeter-based security alone often cannot address harmful actions carried out by authorized users. In contrast to external attackers, insiders act within the scope of authorized access, which makes harmful or risky activity hard to separate from routine, legitimate behavior.

Recent studies increasingly examine behavioral analytics and machine learning methods applied to enterprise activity logs to detect unusual patterns that may indicate insider threats. Many studies have examined unsupervised learning for insider threat detection, mainly because there is little labeled data for insider attacks.

Al-Shehari et al [4], present an anomaly-based model for detecting insider threats that applies the Isolation Forest algorithm. The method learns typical user behavior from features extracted from system logs and flags activities that deviate from that pattern as possible insider threats. The results indicate that Isolation Forest works well with high-dimensional data and imbalanced datasets that often occur in enterprise settings. The study mainly assesses anomaly detection performance and does not address how detected anomalies could be verified or put into practice within access control or incident response processes.

Le and Zincir-Heywood [5] built on this line of work by examining unsupervised, ensemble-based anomaly detection methods for identifying insider threats. The study integrates several unsupervised models to reduce false positives and improve stability when analyzing large enterprise log data. The findings suggest that ensemble methods often perform better than single-model approaches, especially in complex settings where user behavior varies widely. Even with these improvements, the study mainly evaluates detection accuracy and does not clearly explain how to interpret anomaly scores as indicators of risk or how to use them in later security decisions.

Prior studies note that the way data are represented, and the level of detail they capture, can influence insider threat detection outcomes. Le and Zincir-Heywood [6] examined how changes in data granularity affect machine learning approaches to insider threat detection. Their results suggest that combining low-level events into higher-level descriptions of behavior can change detection performance to a meaningful degree. This study supports the use of session-level or aggregated behavior modeling, but it does not provide a clear method for checking whether the detected anomalies align with an organization's security policies.

Alongside traditional anomaly detection models, researchers have examined deep learning methods to represent complex temporal and behavioral patterns. Ma and Rastogi [7] proposed DANTE, an LSTM-based method for predicting insider threats by learning sequential patterns in system logs. Their approach showed that temporal deep learning models can capture long-term dependencies in user behavior. These methods often need large computing resources and complex training, and they are often hard to interpret, which can limit

their use in practical enterprise settings. Survey and review studies place these approaches in context by describing recurring challenges in insider threat detection. Liu et al [8] presented a survey of methods for detecting insider threats, including behavioral analysis, anomaly detection, and approaches based on machine learning. The survey identifies several limits in prior studies: false-positive rates are often high, labeled datasets are scarce, and detection results are not clearly tied to security controls that can be implemented. These observations indicate that detection on its own is not enough and should be paired with validation and decision-support tools.

Prior studies suggest that unsupervised and behavior-based machine learning methods can detect unusual insider actions. Most studies focus on detection performance on its own, and give less attention to validating anomalies or examining how they affect access control decisions. This gap motivates the proposed study. It builds on unsupervised anomaly detection and session-level behavioral modeling, and adds policy-based validation with risk-aware interpretation to make insider threat mitigation in enterprise settings more practical and easier to interpret.

III. METHODOLOGY

This study uses a behavior-driven method to estimate security risk in enterprise settings by examining user activity logs and combining anomaly detection with access control reasoning. The method follows a staged pipeline. It starts by converting raw system logs into session-level representations of user behavior, then applies unsupervised learning to flag atypical activity, and finally checks the flagged cases against policy rules to confirm whether they indicate risk. Because labeled attack data are hard to obtain for insider threats, this approach models normal user behavior and flags departures from that baseline as possible indicators of increased security risk [9]. The method has four stages. First, raw logs are cleaned and grouped into user sessions to form the dataset. Second, session-level behavioral features are computed from these sessions. Third, anomalies are identified with an unsupervised Isolation Forest model. Fourth, the results are mapped to risk categories and checked against predefined policies to validate them for use in access control decisions. Each stage is intended to be easy to interpret and suitable for use in real enterprise contexts.

A. Dataset and Log Preparation

We ran the experiments using the CERT Insider Threat Dataset, a standard benchmark that models typical enterprise user activity [10]. The dataset includes activity logs that record day-to-day interactions of several users in an organizational setting. Its structured but realistic design makes it suitable for studying insider threat cases in which harmful or unauthorized actions are mixed into ordinary day-to-day behavior.

This study examines authentication logs, specifically logon records, to analyze patterns in user access activity. CSV files and file activity logs were collected for analysis. Authentication logs record when users sign in and sign out, which supports analysis of access times and common patterns of use. File activity logs track actions on files, including

opening, copying, writing, and deleting, which supports review of how data is handled over time.

We also retain metadata on removable media access because, in enterprise environments, this activity is often linked to higher security risk. Because individual log entries record separate events and provide little context, we group user actions into sessions before conducting analysis. In this study, a session refers to an uninterrupted series of user actions, where a new session begins after more than 30 minutes with no activity. Representing the data as sessions keeps the time order intact, limits noise from one-off events, and provides a clear unit for later extraction of behavioral features [11].

B. Behavioural Feature Extraction

Behavioral feature extraction is the process of turning observed behavior into measurable variables that can be used for analysis or modelling [22]. It involves selecting what actions to record, deciding how to summarize them over time, and translating the raw observations into features such as frequency, duration, sequence patterns, and variability. In research settings, this step requires clear operational definitions, attention to missing or biased observations, and documentation so that other researchers can reproduce the same features from the same data [11,23].

- After sessions are constructed, each session is described with behavioral features that reflect timing patterns, activity level, and actions relevant to security. The aim of feature extraction is to convert raw session activity into numerical features that capture key patterns in user behavior, remain easy to interpret, and can be used in unsupervised analysis.
- Temporal features include the hour a session starts and the session duration, which indicate the timing and length of a user's interaction with the system. These features separate typical activity during normal working hours from atypical access times, such as late-night logins or unusually long sessions.
- Activity volume is measured as the total count of actions within a session, which serves as an indicator of how intense the interaction was. File operation features are obtained by counting, for each session, how many times files are opened, copied, written, and deleted. These operations represent varying degrees of interaction with the data and may involve different levels of sensitivity.
- Indicators tied to the use of removable media are included to capture behavior that is often linked to unauthorized data transfers. An after-hours activity ratio is then calculated to measure the share of actions that take place outside standard working hours. Taken as a set, these features produce a session-level representation that reflects relevant behavior and supports anomaly detection.

C. Unsupervised Anomaly Detection

Unsupervised anomaly detection can be used to flag sessions that differ from normal user behavior patterns, without needing labeled examples of attacks. Because insider threat labels are scarce and often unreliable, this study assumes that most observed sessions reflect normal behavior.

Given this assumption, we apply an Isolation Forest model to learn what typical session-level behavior looks like and to flag patterns that differ from it as anomalous [12]. Before training the model, we standardize all numerical features measured at the session level so that different feature types are on a comparable scale. The Isolation Forest model was set with a fixed number of estimators, and the contamination value was chosen to match the expected share of anomalous sessions in the dataset [25].

During training, the model repeatedly divides the feature space by choosing a feature and a split point at random. Sessions that are anomalous tend to be separated from the rest after only a few splits, while typical sessions usually require more splits to be isolated [21]. In each session, the trained model produces an anomaly score based on the mean path length needed to isolate the relevant feature vector across the tree ensemble. Higher anomaly scores mean that the observed behavior differs more from the patterns that are typical in the dataset [24].

These scores should not be read as direct evidence of malicious activity [13]; instead, they are used as relative measures of how unusual a behavior is. These anomaly scores are the main input for later risk grouping and validation steps in the framework.

D. Risk Categorization and Policy-Based Validation

The Isolation Forest anomaly scores were mapped to clear risk categories so that security staff can use them in decision-making. Because anomaly scores are relative, not absolute, we set the threshold using percentiles so the rule stays consistent across datasets with different score distributions [12,26].

- Sessions with anomaly scores below the 70th percentile are classified as low risk.
- Scores from the 70th to the 95th percentile are classified as medium risk.
- Scores above the 95th percentile are classified as high risk.

This method does not rely on arbitrary score values and still keeps the same ranking of behavioral deviation. To test whether the assigned risk levels are meaningful in practice, we apply a policy-based validation method that checks them against predefined enterprise security rules. These policies describe behaviors that are often treated as higher risk in organizations, such as:

- Accessing or changing sensitive files after normal working hours
- Using removable media
- Making many file changes within a short period

In each session, we assess compliance with these rules and calculate the number of policy violations as a measure of the level of security concern. Next, we examine how the assigned risk levels match observed policy violations to assess whether the proposed risk categories are appropriate. High-risk sessions are expected to contain more policy violations, whereas low-risk sessions should mostly align with organizational norms.

This validation step offers empirical support that the anomaly-based risk scores reflect security-relevant behavior, rather than chance variation [13, 27].

IV. RESULTS AND DISCUSSION

This section reports an experimental evaluation of the framework using session-level behavioral data. The results are grouped to evaluate anomaly detection performance, risk categories, policy-based checks, and how adding behavioral risk changes access control decisions.

A. Dataset and Session Statistics

Table 1(a) summarizes the dataset after preprocessing and the construction of sessions. The final dataset includes many user sessions, built from authentication records and file activity logs collected across multiple users. The dataset size matches typical enterprise settings and includes enough variation in user and system behavior to support the evaluation of anomaly detection and risk assessment methods.

Table 1(b) reports the main statistical characteristics of the sessions that were constructed. The results indicate that most sessions include only a few events and last for a short time, which is consistent with routine user behavior such as brief logins or limited file access. By contrast, a small share of sessions has many more events and lasts longer, which produces a long-tailed pattern in session behavior.

This kind of variation is typical in organizations, where day-to-day patterns are common and periods of high or unusual activity occur only on occasion. This distribution aligns with the assumptions used in unsupervised anomaly detection and indicates that the session construction process retains meaningful behavior patterns for later analysis.

Table 1(a): Dataset Overview After Preprocessing

	Value
Total events	5545168
Total sessions	300000
Total users	4000

Table 1(b): Session-Level Statistical Summary

	Event count	Session_Duration_Min
Count	300000.000000	300000.000000
Mean	1.448327	2.322226
Std	1.918886	8.947139
Min	1.000000	0.000000
25%	1.000000	0.000000
50%	1.000000	0.000000
75%	1.000000	0.000000
Max	60.000000	226.750000

B. Anomaly Detection Results

Table 2a summarizes the statistical properties of the anomaly scores generated by the Isolation Forest model for all sessions. The scores cluster within a mid-range, while a smaller number of sessions show clearly higher anomaly values. The pattern indicates that most sessions align with expected behavior, whereas a small share departs from the usual activity pattern.

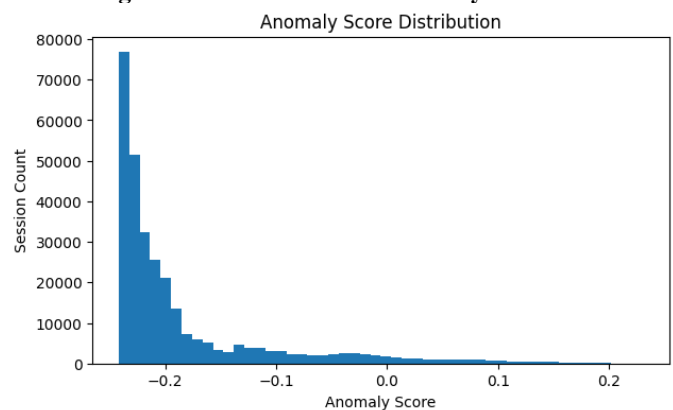
Figure 1 shows how anomaly scores are distributed across the full dataset. The distribution has a long-tail pattern, which suggests that rare cases include large deviations in behavior. This distribution fits unsupervised anomaly detection in enterprise settings, since atypical behavior tends to occur rarely compared with routine activity [14]. The approach assigns risk labels cautiously, which limits false alarms while still maintaining sensitivity to unusual behavioral patterns.

These anomaly scores form the basis for later risk classification and for validation against policy rules.

Table 2(a): Anomaly Score Statistics

	Anomaly_Score
Count	300000.000000
Mean	-0.184179
Std	0.079266
Min	-0.242001
25%	-0.232770
50%	-0.219002
75%	-0.179705
Max	0.230449

Figure 1: Distribution of Anomaly Scores



C. Risk Categorization and Distribution

Raw anomaly scores were translated into practical security interpretations by assigning risk categories based on their percentile ranks. We grouped sessions into low-, medium-, and high-risk categories using their anomaly scores, and we labeled only the largest deviations from typical behavior as high risk. This approach assigns risk labels cautiously, which limits false alarms while still maintaining sensitivity to unusual behavior [15].

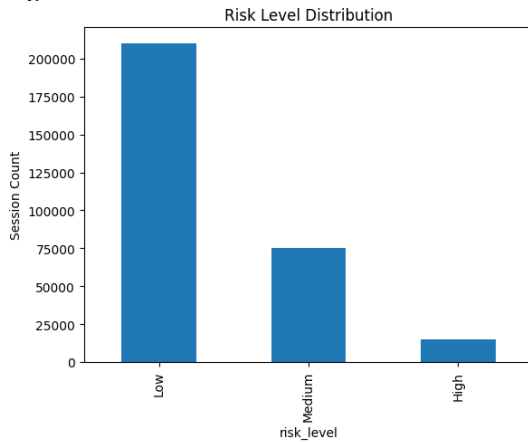
Table 3a reports how sessions are distributed across the three risk categories. The results indicate that most sessions are classified as low risk, with a smaller share classified as medium risk. Only a small share of sessions are labeled high risk, consistent with the assumption that truly risky behavior is uncommon in enterprise settings.

Figure 2 shows the distribution of sessions across risk levels. The clear separation between categories shows that the thresholding strategy turns continuous anomaly scores into risk labels that are easy to interpret. Limiting the high-risk category to a small set of sessions makes later security decisions more reliable and provides a controlled input for policy-based validation and access control assessment.

Table 3(a): Risk Level Distribution

Risk level	Proportion
Low	70.003333
Medium	24.997000
High	4.999667

Figure 2: Risk-Level Distribution of User Sessions



D. Policy-Based Validation of Risk Levels

To assess whether the assigned risk levels reflect real security concerns, we conducted a policy-based validation using predefined enterprise security rules. These rules describe behaviors that are often treated as higher risk in security monitoring, such as access to sensitive files outside normal working hours, use of removable storage, and short periods with many file changes [15].

Table 4a reports the mean number of policy violations in each risk category, and **Table 4b** reports the share of sessions with at least one policy violation.

The results indicate that most violations occur in the high-risk category, while low-risk sessions show few or no violations.

Figure 3 also shows this separation, suggesting that the risk categories track security-related behavior rather than chance variation.

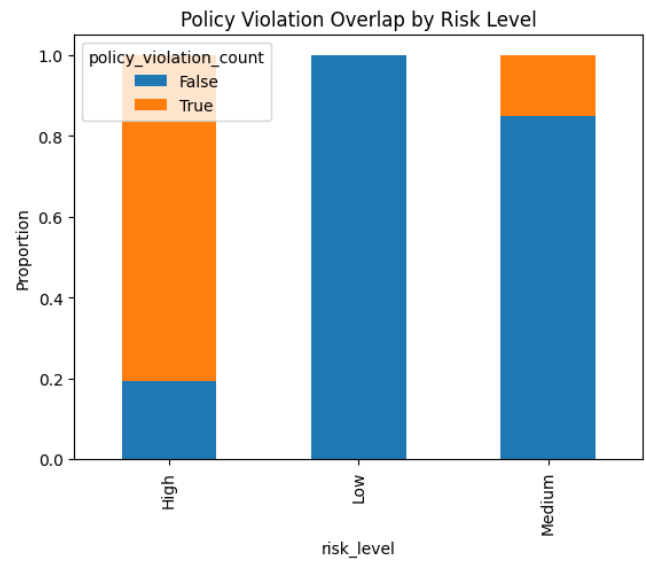
Table 4(a): Average Policy Violations per Risk Level

Risk Level	Policy Violation Count
High	1.429695
Low	0.000000
Medium	0.161753

Table 4(b): Percentage of Sessions with Policy Violations

Risk Level	Any Violation
High	80.538703
Low	0.000000
Medium	15.140484

Figure 3: Policy Violation Overlap by Risk Level



E. Access Control Impact Analysis

To evaluate the practical effect of adding behavioral risk to authorization decisions, we compared a static role-based access control (RBAC) model with a risk-aware access control approach. **Table 5a** presents the number of sessions that were restricted under each strategy. Static RBAC does not limit sessions based on user behavior [16], whereas the risk-aware approach restricts only those sessions that are classified as high risk. These findings show that adding behavioral risk supports targeted interventions without imposing blanket limits on normal user activity, which supports the case for risk-aware access control in enterprise settings [17].

Table 5(a): Static RBAC vs Risk-Based Access Control Decisions

	Approach	Restricted Sessions
0	Static RBAC	0
1	Risk-based RBAC	14999

V. CONCLUSION AND FUTURE SCOPE

This study proposes a behavior-driven risk assessment framework to improve access control decision-making in enterprise settings. Grouping raw authentication and file activity logs into session-level records preserves behavioral context that event-by-event analysis often overlooks. An unsupervised Isolation Forest model was used to flag unusual sessions without labeled insider threat data, which addresses a practical constraint common in real security work.

The experimental results show that the proposed risk categorization distinguishes routine sessions from sessions with atypical behavior. Policy-based validation indicated that high-risk sessions largely corresponded to predefined security rule violations, while low-risk sessions showed few or no violations. This validation offers empirical support that the anomaly-based risk scores align with security-relevant behavior rather than chance variation.

A comparative evaluation showed the limits of static role-based access control. When behavioral risk is included, access can be restricted for high-risk sessions while routine user activity remains largely unaffected. The proposed framework does not claim to identify malicious users with certainty. Instead, it provides a practical and interpretable way to rank sessions and users that may require closer review or tighter access controls.

In this sense, it marks a move toward access control systems that can adapt to changing conditions and account for situational context [18]. Future research could study temporal deep learning models that track long-term behavior over multiple sessions [19], and it could also test broader inputs, including network traces and application-level logs. Future work should extend the framework so it can run in near real time and assess its performance in live enterprise settings [20].

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